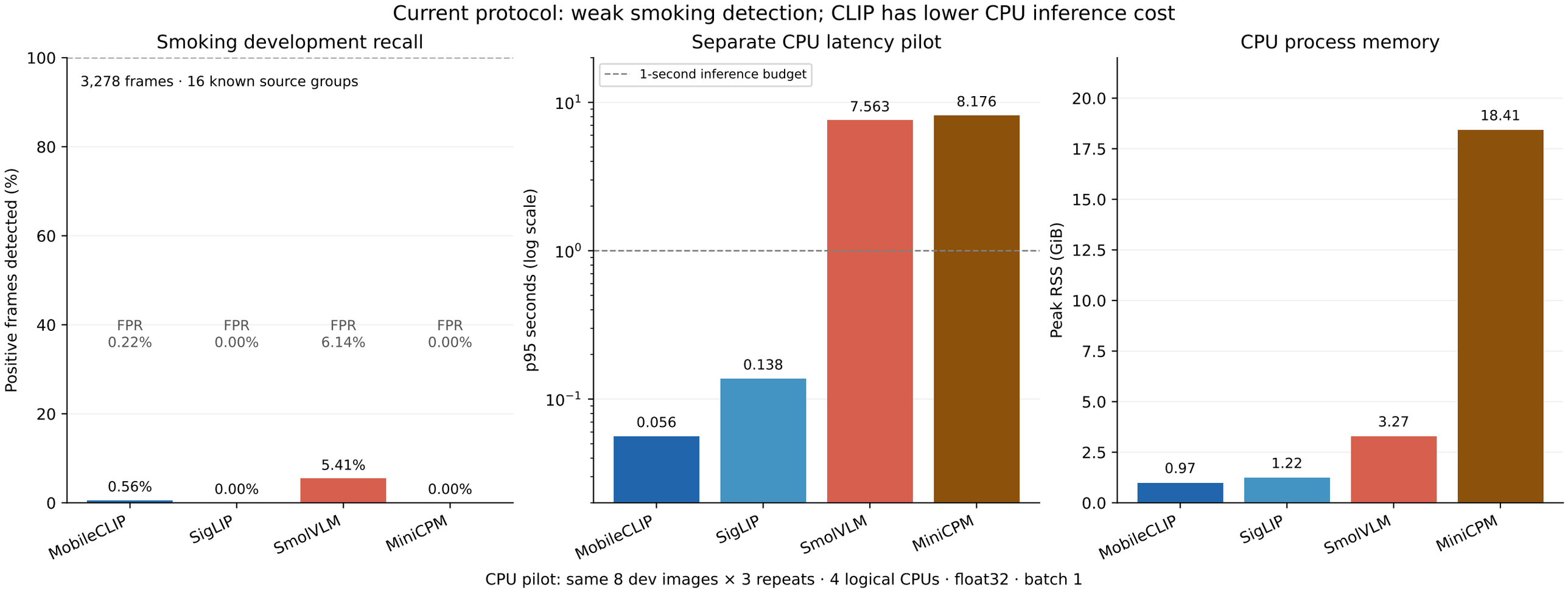
Current protocol: weak smoking detection; CLIP has lower CPU inference cost
Smoking development recall
3,278 frames · 16 known source groups
Positive frames detected (%)
100
80
60
40
20
0
FPR 0.22%
FPR 0.00%
FPR 6.14%
FPR 0.00%
0.56%
0.00%
5.41%
0.00%
MobileCLIP
SigLIP
SmolVLM
MiniCPM
Separate CPU latency pilot
1-second inference budget
p95 seconds (log scale)
10^{1}
10^{0}
10^{-1}
0.056
0.138
7.563
8.176
MobileCLIP
SigLIP
SmolVLM
MiniCPM
CPU process memory
Peak RSS (GiB)
20.0
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0
0.97
1.22
3.27
18.41
MobileCLIP
SigLIP
SmolVLM
MiniCPM
CPU pilot: same 8 dev images × 3 repeats · 4 logical CPUs · float32 · batch 1